

- 12** A certain star of mass M and radius r rotates so rapidly that material at its equator only just remains on its surface. Given that the gravitational constant is G , the period of rotation is

A $2\pi\sqrt{\frac{r}{G}}$

B $2\pi\sqrt{\frac{G}{r}}$

C $2\pi\sqrt{\frac{r}{MG}}$

D $2\pi\sqrt{\frac{r^3}{MG}}$

Ans: D

$$\frac{GMm}{r^2} = mr \left(\frac{2\pi}{T} \right)^2$$

$$T^2 = r^3 \frac{4\pi^2}{GM}$$